Indian Institute of Technology Kanpur
Department of Mathematics and Statistics
Analysis I (MTH 301)
Mid-semester examination

Date: 21/09/2024

Time: 2 hours | Total marks: 35

Solutions

1. Let $\{x_n\}_{n=1}^{\infty}$ be a sequence of nonnegative real numbers satisfying the following:

$$x_{n+1} \leqslant x_n + \frac{1}{n^2}, \text{ for all } n \in \mathbb{N}. \tag{1.1}$$

Show that $\{x_n\}_{n=1}^{\infty}$ is convergent. [5]

Solution. *Let $\varepsilon > 0$. Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent, there exists $N \in \mathbb{N}$ such that, for all $n, m \geqslant N$ with $m > n$ one has $\sum_{k=n+1}^{m} \frac{1}{k^2} < \varepsilon$. Let $n \geqslant N+1$. A repeated application of (1.1) yields that*

$$\begin{aligned} x_m &\leqslant x_{m-1} + \frac{1}{(m-1)^2} \leqslant x_{m-2} + \frac{1}{(m-2)^2} + \frac{1}{(m-1)^2} \\ &\leqslant \\ &\vdots \\ &\leqslant x_n + \left(\frac{1}{n^2} + \cdots + \frac{1}{(m-1)^2}\right) < x_n + \varepsilon, \ \forall m \geqslant n. \end{aligned}$$

This shows that $\limsup_{n\to\infty} x_n \leqslant \sup_{m\geqslant n} x_m \leqslant x_n + \varepsilon$, whenever $n \geqslant N+1$. As all x_n's are nonnegative, so is $\limsup_{n\to\infty} x_n$. Hence $\inf_{n\geqslant N+1} x_n$ exists finitely and it satisfies

$$\limsup_{n\to\infty} x_n \leqslant \left(\inf_{n\geqslant N+1} x_n\right) + \varepsilon \leqslant \liminf_{n\to\infty} x_n + \varepsilon.$$

Since ε is arbitrary, one concludes $\limsup_{n\to\infty} x_n \leqslant \liminf_{n\to\infty} x_n$. The convergence of $\{x_n\}_{n=1}^{\infty}$ is now immediate from this as one always has $\limsup_{n\to\infty} x_n \geqslant \liminf_{n\to\infty} x_n$.

Common mistakes.

1. *While it is true that, by repeated application of (1.1) that, $x_m - x_n \leqslant \frac{1}{n^2} + \cdots + \frac{1}{(m-1)^2}$, but this does not imply that $|x_m - x_n| \leqslant \frac{1}{n^2} + \cdots + \frac{1}{(m-1)^2}$. In other words, taking modulus of both sides of an inequality does not preserve the same, for instance $-5 \leqslant 2$ but $5 > 2$.*
2. *$\lim_{n\to\infty} x_n$ does not make sense until it is established that $\{x_n\}_{n=1}^{\infty}$ is convergent. So taking limits of both sides of (1.1) right away is not permitted.*

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3. *$\{x_n\}_{n=1}^{\infty}$ need not be monotone. It can be oscillatory as well. In fact, $\left\{1+\frac{(-1)^n}{2n^2}\right\}_{n=1}^{\infty}$ is an example of a sequence satisfying (1.1).*

4. *As $\{x_n\}_{n=1}^{\infty}$ need not be monotone, showing $\{x_n\}_{n=1}^{\infty}$ is bounded does not help.*

5. *Some students have argues like the following: since $x_n \leqslant x_1 + \sum_{k=1}^{n-1} \frac{1}{k^2}$, from comparison test we get $\{x_n\}_{n=1}^{\infty}$ is convergent. This is wrong as there is no such comparison test for convergence of a sequence.*

6. *$x_{n+1} - x_n \leqslant \frac{1}{n^2}$, for all $n \in \mathbb{N}$. But as $\{x_n\}_{n=1}^{\infty}$ need not be monotone, you can't conclude that $\sum_{n=1}^{\infty}(x_{n+1} - x_n)$ is convergent.*

7. *Some of you have assumed that $\{x_n\}_{n=1}^{\infty}$ is Cauchy beforehand. Of course, this is not given; in fact, this is equivalent to what you are supposed to show.*

2. Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \longrightarrow Y$ be continuous and onto. Assume that

$$d_X(x, y) \leqslant d_Y(f(x), f(y)), \text{ for all } x, y \in X.$$

Determine whether each of the following statements is true or false:

(I) If (X, d_X) is complete, then (Y, d_Y) is complete.

(II) If (Y, d_Y) is complete, then (X, d_X) is complete.

Justify your answer. [5]

Solution. *(I) is true. Let $\{y_n\}_{n=1}^{\infty}$ be a Cauchy sequence in Y. Suppose that $\varepsilon > 0$. Then there exists $N \in \mathbb{N}$ such that, for all $n, m \geqslant N$, $d(y_n, y_m) < \varepsilon$. Since f is onto, for all $n \in \mathbb{N}$, there exists x_n such that $y_n = f(x_n)$. Then it is clear that, for all $n, m \geqslant N$,*

$$d(x_n, x_m) \leqslant d(f(x_n), f(y_n)) \leqslant d(y_n, y_m) < \varepsilon.$$

This shows that $\{x_n\}$ is Cauchy. As X is complete, there exists $x \in X$ such that $x_n \xrightarrow[n\to\infty]{} x$. From the continuity of f, we obtain that $y_n = f(x_n) \xrightarrow[n\to\infty]{} f(x)$.

(II) is FALSE. Let $(0, 1]$ and $[1, \infty)$ both be equipped with the metrics induced by the usual metric on $\mathbb{R}$, and $f : (0, 1] \longrightarrow [1, \infty)$, $f(x) \stackrel{\text{def}}{=} \frac{1}{x}$. Clearly f is continuous and onto, and further to that

$$|f(x) - f(y)| = \left|\frac{1}{x} - \frac{1}{y}\right| = \frac{|x - y|}{xy} \geqslant |x - y|,$$

as $0 < x, y \leqslant 1$. Now $(0, 1]$ is not complete but $[1, \infty)$ is complete being a closed subset of the complete metric space $\mathbb{R}$.

Common mistakes.

1. *"False" is true and "True" is false.*

2. *$f : (0, 1] \longrightarrow \mathbb{R}$, $f(x) \stackrel{\text{def}}{=} \frac{1}{x}$ is not onto. Therefore this does NOT serve as a counter example for (II).*

3. *It is NOT true that a continuous function always sends Cauchy sequences to Cauchy sequence. For instance, the map mentioned above in 2, sends the Cauchy sequence* $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ *to an unbounded sequence.*

4. *Some students have started with a convergent sequence* $a_n \xrightarrow[n\to\infty]{} a$ *in* X *and showed that* $\{f(a_n)\}_{n=1}^{\infty}$ *is convergent. This is just a restatement of continuity of* f*, and does NOT show that an arbitrary Cauchy sequence in* Y *converges.*

5. *It does not make any sense when you say a function from a metric space to another metric space is monotone.*

3. Let $f : (0,1] \longrightarrow \mathbb{R}$ be a continuous function such that $\lim\limits_{x\to 0+} f(x)$ exists in $\mathbb{R}$. Does f necessarily have to be uniformly continuous? Justify your answer. [3]

Solution. *Define* $\tilde{f} : [0,1] \longrightarrow \mathbb{R}$ *by*

$$\tilde{f}(x) \stackrel{\text{def}}{=} \begin{cases} f(x), & \text{if } 0 < x \leqslant 1; \\ \lim\limits_{x\to 0+} f(x), & \text{if } x = 0. \end{cases}$$

Clearly $\tilde{f}$ is contunous. Since $[0,1]$ is compact, $\tilde{f}$ is uniformly continuous. Consequently its restriction to $(0,1]$, which is f, is also unifomly continuous.

Common mistakes.

1. *Some students have considered the example* $f : (0,1] \longrightarrow \mathbb{R}$, $f(x) \stackrel{\text{def}}{=} \frac{1}{x}$. *But this does not help as* $\lim\limits_{x\to 0+} f(x) = +\infty$ *does not exist in* $\mathbb{R}$.

2. $f : (0,1] \longrightarrow \mathbb{R}$, $f(x) \stackrel{\text{def}}{=} x^2$ *is uniformly continuous. In fact, it is uniformly continuous on any subinterval of* $\mathbb{R}$ *having an upper bound, but not uniformly continuous on intervals of the form* (α, ∞) *or* $[\alpha, \infty)$.

3. *A function may not be uniformly continuous despite being bounded.*

4. Prove or disprove the following statement: if $\{A_n\}_{n=1}^{\infty}$ is a sequence of subsets of $\mathbb{R}$ such that $\mathbb{R}\backslash\mathbb{Q} = \bigcup\limits_{n=1}^{\infty} A_n$, then the closure of some A_n must contain an open interval. [3]

Solution. *Let* $B \stackrel{\text{def}}{=} \bigcup_{n=1}^{\infty} \overline{A_n}$. *Clearly* $\mathbb{R}\backslash B \subseteqq \mathbb{Q}$, *and hence it is countable. But as*

$$\mathbb{R}\backslash B = \bigcap_{n=1}^{\infty} \left(\mathbb{R}\backslash\overline{A_n}\right),$$

there exists $n \in \mathbb{N}$ *such that the open set* $\mathbb{R}\backslash\overline{A_n}$ *is not dense. Consequently, that* $\overline{A_n}$ *must have nonempty interior.*

Common mistakes.

1. *Infinite union of closed sets need NOT be closed. Such examples have already appeared in our course.*
2. *Intersection of dense sets need not be dense; it can be even empty.*
3. *It is NOT true that* $\overline{\bigcup_{n=1}^{\infty} A_n} = \bigcup_{n=1}^{\infty} \overline{A_n}$. *Example:* $\bigcup_{x\in\mathbb{Q}}\{x\} = \mathbb{Q} \subsetneq \overline{\mathbb{Q}} = \mathbb{R}$.
4. *It is NOT true that a countable union of nowhere dense subsets will always be nowhere dense. An immediate example is* $\mathbb{Q} = \bigcup_{n=1}^{\infty}\{x\}$.
5. $\mathbb{Q}$ *is neither open nor closed in* $\mathbb{R}$. *Consequently, the same applies to* $\mathbb{R}\backslash\mathbb{Q}$.
6. $\mathbb{R}\backslash\mathbb{Q}$ *is NOT nowhere dense.*
7. *It has not been assumed that any of the* A_n*'s is closed. So there is no guaranty that any* $X\backslash A_n$ *will be open.*
8. *Similar to the above, it has not been assumed anywhere that* A_n*'s are open.*
9. *There are subsets of the real line that are neither open nor closed. So it is NOT right to conclude that if a set is not closed then it is open.*
10. *Not nowwhere dense does not imply that the set would be dense. For example,* $[0,1]$ *is neither nowhere dense nor dense in* $\mathbb{R}$.
11. *Some students assumed right in the beginning that some open interval* $(a,b) \subseteqq \mathbb{R}\backslash\mathbb{Q}$. *Isn't this an impossibility?*

5. Let $n \geqslant 2$ be an integer. Assume that $M_n(\mathbb{R})$ is equipped with the *sup norm*, i.e.,

$$\|A\|_\infty \stackrel{\text{def}}{=} \max_{1\leqslant i,j\leqslant n} |a_{ij}|, \text{ where } A = (a_{ij})_{i,j}.$$

Consider the set $\mathcal{S} \stackrel{\text{def}}{=} \{A \in M_n(\mathbb{R}) : \operatorname{rank} A \leqslant n-1\}$. Which of the following statements about $\mathcal{S}$ is/are true:

(A) $\mathcal{S}$ is open. (B) $\mathcal{S}$ is closed. (C) $\mathcal{S}$ is nowhere dense.

Justify your answer. [3]

Solution. *Observe that* $\mathcal{S} = \{A \in M_n(\mathbb{R}) : \det A = 0\}$. *The complement of* $\mathcal{S}$ *is* $\mathrm{GL}_n(\mathbb{R})$ *which is open and dense. So* $\mathcal{S}$ *is closed and nowhere dense. Finally* $\mathcal{S}$ *cannot be open as every open ball has nonempty intersection with its complement. So the correct options are precisely (B) and (C).*

Note. *The reason for* $\mathcal{S}$ *not being open is* $M_n(\mathbb{R})\backslash\mathcal{S}$ *is dense in* $M_n(\mathbb{R})$. *In a metric space a subset can be open despite its complement being so. Hence this is not a valid reason that* $\mathcal{S}$ *is not open as its complement is open.*

Common mistakes.

1. *Some of you have considered the function* $f : M_n(\mathbb{R}) \longrightarrow \mathbb{R}$, $f(A) \stackrel{\text{def}}{=} \operatorname{rank} A$. *This function is NOT continuous.*

2. *Some of the students have argued correctly that (B) and (C) are right, but did not provide any reason why (A) cannot be true. So until you rule out (A), the answer is NOT complete.*

6. Let $C[0,1]$ denote the set of all real valued continuous functions on $[0,1]$. For $f \in C[0,1]$, define

$$Tf : [0,1] \longrightarrow \mathbb{R},\ Tf(x) \stackrel{\text{def}}{=} \int_0^x f(t)dt.$$

Find all fixed points for $T : C[0,1] \longrightarrow C[0,1]$ using the **contraction principle** for complete metric spaces. [3]

Solution. *Let $C[0,1]$ be equipped with the sup norm, which is a complete metric space. It is easy to verify that T^2 is a contraction as for any $f \in C[0,1]$, one has the following:*

$$\begin{aligned}
|T^2 f(x)| = |T(Tf)(x)| = \left|\int_0^x Tf(t)dt\right| &= \left|\int_0^x \left(\int_0^t f(s)ds\right) dt\right| \\
&\leqslant \int_0^x \left|\int_0^t f(s)ds\right| dt \\
&\leqslant \int_0^x \left(\int_0^t |f(s)|ds\right) dt \\
&\leqslant \int_0^x (\|f\|_\infty)\, t\, dt \\
&\leqslant \|f\|_\infty \int_0^x t\, dt \\
&\leqslant \|f\|_\infty \left(\frac{x^2}{2}\right) \\
&\leqslant \frac{\|f\|_\infty}{2}, \text{ for all } x \in [0,1].
\end{aligned}$$

Hence T has a unique fixed point. Since it is readily seen that the constant zero function is a fixed point for T, it has to be that unique fixed point.

Common mistakes.

1. *Choice of the norm on $C[0,1]$ must be correct.*
2. *T is NOT a contraction. Indeed, if $\mathbf{1}$ denotes the constant function whose value is 1 everywhere, then $T\mathbf{1}$ is the identity function on $[0,1]$, so that $\|T\mathbf{1}\|_\infty = \sup_{x\in[0,1]} x = 1$.*

   Hence no marks has been awarded if your argument is based on the void claim that T is a contraction.
3. *The exponential function on $[0,1]$, i.e., e^x, is not a fixed point for T.*
4. *Some students have written things like "$x = 0$ is a fixed point" etc.*
5. *Some of you have started differentiating the functions. Note that the functions are assumed to be continuous only.*

6. *Quite a few students have argued as follows: if $f \in C[0,1]$ is a fixed point for T, then $f'(x) = f(x)$, for all $x \in [0,1]$. Let $y = f(x)$. Then*

$$\begin{aligned} &\frac{dy}{y} = dx \qquad (6.2)\\ \Rightarrow &\int \frac{dy}{y} = \int dx\\ \Rightarrow &\log y = x + c\\ \Rightarrow &y = Ae^x. \end{aligned}$$

Since $f(0) = 0$, so $A = 0$, which implies $y = 0$. This is argument is self-contradictory, as (6.2) assumes that y cannot vanish anywhere, while the final conclusion made is $y = 0$.

7. Prove that if a subset of $\mathbb{R}$ has no limit point in $\mathbb{R}$ then it must be countable. [5]

Solution. *Assume the contrary, i.e, A is uncountable. It is easy to see that there exists a closed and bounded interval that contains infinitely many points of A. This can be done in many ways, for instance, as $\mathbb{R} = \bigcup_{n=1}^{\infty}[-n,n]$, one has $A = \bigcup_{n=1}^{\infty}(A \cap [-n,n])$. Hence there must be some $n \in \mathbb{N}$ such that $[-n,n] \cap A$ is infinite. Now from the compactness of $[-n,n]$, $[-n,n] \cap A$ has a limit point in $[-n,n]$. Clearly this limit point is a limit point of A in $\mathbb{R}$, which is a contradiction.*

Common mistakes.

1. *No credit will be given for writing the definition of an isolated point.*
2. *It is true that for every $x \in A$, there exists $\varepsilon_x > 0$ such that $(x - \varepsilon_x, x + \varepsilon_x) \cap A) = \{x\}$, since all points of A are isolated, however two such open interval, say $(x - \varepsilon_x, x + \varepsilon_x)$ and $(y - \varepsilon_y, y + \varepsilon_y)$, where $x \neq y$, may have nonempty intersection. Hence if we choose a rational number r_x from each $(x - \varepsilon_x, x + \varepsilon_x)$, then r_x and r_y can be same for $x \neq y$.*
3. *Further to that, ε_x may change with x. In fact, for any interval $[n, n+1]$ such that $[n, n+1] \cap A$ is infinite, there is no guaranty that $\min_{x \in [n,n+1] \cap A} \varepsilon_x > 0$.*
4. *Having an injective map from $\mathbb{Q}$ to A does NOT show that A is countable.*
5. *Some of you have said that it suffices to show that the set of isolated points of $\mathbb{R}$ is countable. While this statement is trivial, as it is just empty, it does NOT yield the proof of countability of A.*

8. Recall that, for $(a,b) \in \mathbb{R}^2$ and $r > 0$, the circle with center (a,b) and radius r is defined to be

$$\{(x,y) \in \mathbb{R}^2 : (x-a)^2 + (y-b)^2 = r^2\}.$$

Show that $\mathbb{R}^2$ cannot be expressed as the disjoint union of circles, i.e., there is no collection $\{C_i : i \in I\}$ of circles in $\mathbb{R}^2$ satisfying both the properties:

(i) $C_i \cap C_j = \varnothing$, for all $i, j \in I$ with $i \neq j$,

(ii) $\bigcup_{i \in I} C_i = \mathbb{R}^2$.

Hint. *If C and C' are two disjoint circles such that the latter passes through the center of the former, then how are the radii of C and C' related?*

[8]

Solution. *If possible let $\mathscr{C} = \{C_i : i \in I\}$ be a collection of circles in $\mathbb{R}^2$ satisfying (i) and (ii). Choose a circle, say C_1 in $\mathscr{C}$, and denote its center and radius by $\mathbf{x}_1$ and $r_1 > 0$ respectively. Then there exists a unique circle in $\mathscr{C}$ which passes through $\mathbf{x}_1$. Denote this circle by C_2, and its center and radius by $\mathbf{x}_2$ and r_2 respectively. Since C_1 and C_2 are non-intersecting, we must have $r_2 = \|\mathbf{x}_2 - \mathbf{x}_1\| < \frac{r_1}{2}$. Next pick the unique circle in $\mathscr{C}$, denoted by C_3, for which $\mathbf{x}_2 \in C_3$. If $\mathbf{x}_3$ and r_3 denote its center and radius respectively, then likewise the previous case, we see that $r_3 = \|\mathbf{x}_3 - \mathbf{x}_2\| < \frac{r_2}{2}$. Continuing this way, for each $n \in \mathbb{N}$, we obtain a circle, say C_n, whose center and radius are denoted by $\mathbf{x}_n$ and r_n respectively, such that $r_{n+1} = \|\mathbf{x}_{n+1} - \mathbf{x}_n\| < \frac{r_n}{2}$, $\forall n \in \mathbb{N}$. This shows that*

$$B[\mathbf{x}_{n+1}; r_{n+1}] \subseteqq B(\mathbf{x}_n; r_n), \ \forall n \in \mathbb{N}. \tag{8.3}$$

We now consider the sequence of closed balls $\{B[\mathbf{x}_n; r_n]\}_{n=1}^{\infty}$. From the construction, it is clear that $\operatorname{diam} B[\mathbf{x}_n; r_n] < \frac{r_1}{2^{n-1}} \xrightarrow[n\to\infty]{} 0$. Using Cantor's intersection theorem, we obtain a unique point $\mathbf{x} \in \mathbb{R}^2$ such that $\mathbf{x} \in B[\mathbf{x}_n; r_n]$, for all $n \in \mathbb{N}$. From (8.3), we see that $\mathbf{x} \in B(\mathbf{x}_n; r_n)$, for any $n \in \mathbb{N}$. If $C \in \mathscr{C}$ is the circle that passes through $\mathbf{x}$, then C must be contained in every $B[\mathbf{x}_n; r_n]$, which is absurd.

Common mistakes.

1. *If C and C' are two disjoint circles such that the latter passes through the center of the former, then C' lie in the open ball whose boundary is C. Furthermore, radius of C' is less than $\frac{1}{2}\times$ radius of C. Those who have considered C and C' are intersecting or C' lies "outside" C are completely wrong.*
2. *I has not been assumed to be countable. Hence Baire category theorem does not apply here, as at the most it only shows that $\mathbb{R}^2$ cannot be written as the union of countably many circles.*
3. *Further to 2, in fact, one does not require Baire category theorem at all to show that $\mathbb{R}^2$ cannot be a countable union of disjoint circles, as the intersection of each such circle and the x-axis contains at most two points. So using the Baire category theorem for that is a complete overkill.*
4. *Some of you considered a sequence $\{C_n\}_{n=1}^{\infty}$ of circles in $\{C_i : i \in I\}$ with the property that each C_{i+1} lies inside the open ball whose boundary is C_i and radius of C_i goes to zero. Cantor's intersection theorem cannot be applied to this sequence as C_{i+1} is not a subset of C_i.*
5. *I have also seen the following argument: since an open interval $(a,b) \subseteqq \mathbb{R}^2 = \bigcup_{i\in I} C_i$, so $(a,b) \subseteqq C_i$, for some $i \in I$. This need not be true even if the collection is countable, as (a,b) is not an open subset of $\mathbb{R}^2$.*
6. *A few students have applied Cantor's intersection theorem to the correct sequence but did not provide any justification why the point in the intersection cannot belong to any C_i (note that you have to show that, for ALL $i \in I$, C_i cannot contain that point).*